

MOTIONCORE · REQUIREMENTS

MotionCore requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/motioncore		
Source	docs/03-requirements.md @ 1757f3f		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R1 upper bound 58 to 60 V; R12 mass limit 1.5 to 2.0 kg; status from MTC-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3. R2 heavy case relaxed to 350 W and R14 redefined to the MotionCore kit without the motor (MTC-DDR-001, adopted for TRL 3 pending Amish's review); status column from MTC-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

MotionCore requirements

These requirements are checked by calculation in MTC-CAL-001 v0.2. Four targets have changed since TRL 2, all decided by Amish on 2026-09-25 (MTC-DDR-001 and MTC-DDR-002): the heavy case in R2 is 350 W instead of 500 W; R14 covers the MotionCore kit without the reference motor; the upper bound of R1 is 60 V instead of 58 V, so that CellGuard's 16S LiFePO4 packs are covered; and the mass limit in R12 is 2.0 kg instead of 1.5 kg, because the enclosure is also the heat sink. No requirement has been verified by test.

The **reference case** used throughout is a 250 W geared hub motor on a 48 V (46.8 V nominal) SwapCell pack at 40 °C ambient. The **heavy case** is 350 W continuous on a 25.6 V (8S) LiFePO4 pack at 40 °C ambient.

On paper, ten requirements are met and none is not met. R2, R9 and R10 are **at risk**; R11 cannot be verified at TRL 3. R12 is met with a margin of only 0.06 kg.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (MTC-CAL-001)
R1	Supply range	Operate from 20 to 60 V DC (8S LiFePO4 to 13S lithium-ion and 16S LiFePO4, including SwapCell at 54.6 V full and CellGuard 16S at 58.4 V full); withstand 75 V transients from regeneration	Controller and contactor datasheets; bus voltage calculation	Met (design review): 75 V controller, 18 to 75 V auxiliary buck, bus peaks near 67.8 V with a 66 V controller overvoltage fault
R2	Motor power	250 W continuous in the reference case; 350 W continuous in the heavy case; 750 W for 10 s	Current and thermal calculation	At risk: met electrically (6.9 A, 17.7 A; 750 W peak 21.9 A on SwapCell, 40.9 A on 8S LiFePO4); heavy case limited by R9
R3	Hardwired emergency stop	Twin-channel normally closed e-stop wired in series with the contactor coil; motor power removed within 100 ms of actuation; works with the supervisor and controller both failed	Circuit review; contactor drop-out time from datasheet	Met on paper: 67 ms worst case, 30 ms modeled; unverified
R4	No automatic restart	After an e-stop, power loss or fault, the drive stays off until the e-stop is released, the throttle is at zero and the operator cycles the enable key	Circuit and state-machine review	Met (design review); unverified
R5	Brake interlock	Two brake switch inputs; drive torque removed within 200 ms of either switch opening; a spring-applied brake output released only while the safety loop is closed	Circuit review; supervisor timing budget	Met on paper: about 75 ms; unverified
R6	Independent speed limit	Speed limit set in the controller and checked by the supervisor from a separate sensor; overspeed of more than 10 % for more than 0.5 s opens the contactor; defaults 25 km/h (pedal assist) and 1.5 m/s (walk-behind)	Firmware review; sensor resolution calculation	Met on paper: period timing, fivefold margin over magnet jitter, trip in 0.60 to 0.70 s; unverified
R7	Fault reaction	Supervisor removes torque within 200 ms of: command timeout, throttle signal outside 0.5 to 4.5 V, loss of controller CAN heartbeat, over-temperature or BMS fault flag	Fault table review	Met on paper: about 170 ms worst (controller heartbeat), about 80 ms for a BMS fault; unverified
R8	Precharge	Inrush at contactor closure under 5 A; bus ready within 1 s	Precharge calculation	Met: 0.55 A peak, 1.3 A at closure, ready in 0.6 s
R9	Thermal	Enclosure surface 60 °C or less at 40 °C ambient in natural convection	Thermal calculation	Reference case met (50.2 °C); heavy case at risk (58.5 °C; 65 to 71 °C in sensitivity cases)
R10	Ingress and vibration	Enclosure and mated connectors IP65; survive road and stair vibration without loosening	Datasheets and design review	At risk: the XT90 power socket is not sealed; a sealed power connector is to be evaluated before interface v0.1 is frozen

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (MTC-CAL-001)
				(MTC-DDR-002); vibration not analyzed
R11	One interface	One documented connector set (power in, motor, sensors, safety loop, command) keyed against mis-mating; fitted to a new host in 2 h or less with hand tools	Interface document; timed fit at TRL 4	Not verifiable at TRL 3: interface v0.1 adopted and drawn (MTC-DWG-001)
R12	Size and mass	Module within 250 x 170 x 70 mm; module mass 2.0 kg or less (1.5 kg before MTC-DDR-002)	Model and mass estimate	Met: 243 x 168 x 66 mm; about 1.94 kg, 0.06 kg margin
R13	Open and repairable	Hardware under CERN-OHL-S-2.0; supervisor firmware MIT; controller firmware open source; fault log readable over USB without closed software	License review	Met (design review): separate programs on separate processors, CAN link only; formal license review open
R14	Cost	MotionCore kit (module and devices, BOM items 2 to 14) \$300 or less in parts; the reference motor is costed to each host	Priced BOM (bom/ bom.csv)	Met on indicative prices: \$265 (reference motor \$70 extra)

■ Assumptions

- Motor efficiency about 80 % at rated load; controller losses from a component model (MTC-CAL-001 Table 1), which gives about 98 to 99 % controller efficiency; to be checked against datasheets once parts are chosen.
- Contactor release time 50 ms or less with a 24 V Zener coil suppressor; this is a selection criterion for the part.
- Enclosure conductance about 1.17 W/K from simplified natural convection and radiation correlations, clear anodized, shaded (MTC-CAL-001 section 4).
- The host provides mechanical brakes able to stop it with the motor unpowered. MotionCore removes drive power; it does not replace the host's brakes.
- The host carries out its own risk assessment. MotionCore targets are not a claimed ISO 13849 performance level.